

One-day-ahead 1% VaR forecasting on QQQ daily log returns

Comparison of Multiple model classes

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Project Objective

Build and compare a reproducible VaR forecasting framework that combines:

- Rolling out-of-sample forecasts
- Multiple model classes (parametric, semi-parametric, nonparametric, direct quantile)
- Statistical backtesting
- Plot diagnostics

The real goal is not just in-sample fit. The goal is model comparison under realistic tail-risk forecasting conditions.

Data and sample

Dataset:

- QQQ adjusted close
- VIX close

Construction:

- QQQ close $\rightarrow$ daily log returns
- VIX used as exogenous regressor for GARCHX
- Data pipeline starts at 2015-01-01

Current local files indicate:

- Full modeled sample runs roughly from 2015-01-02 to 2026-03-19/2026-03-20 depending on the saved output file
- Initial training fraction: 70%
- Out-of-sample period used in saved backtests: 846 trading days

Model Suite

A. Original pipeline models

- GARCH(1,1) - Gaussian/Student-t
- GARCH(1,1) with FHS - Gaussian/Student-t
- GARCH(1,1) with Bootstrap - Gaussian/Student-t

B. Extension GARCH models

- GARCHX(VIX)
- GJR-GARCH - Gaussian/Student-t

C. Direct Simulation Methods

- CAViaR-SAV
- Historical Simulation

Use Student-t innovations for GARCH and GJR, while the lecture's explicit benchmark example was Gaussian GARCH. For financial returns it is usually better.

Model formulas / interpretation

GARCH(1,1)

Variance recursion:

$$\sigma_t^2 = \omega + a \times \epsilon_{t-1}^2 + b \times \sigma_{t-1}^2$$

VaR computed from one-step conditional volatility and a parametric quantile.

GARCHX(VIX)

Variance recursion:

$$\sigma_t^2 = \omega + a \times \epsilon_{t-1}^2 + b \times \sigma_{t-1}^2 + g \times X_{t-1}$$

where X_t is a transformed VIX regressor.

Current implementation uses:

GJR-GARCH

Variance recursion:

Backtesting framework

At each out-of-sample date: 1. Fit model on data up to time $t-1$ 2. Forecast one-step VaR for time t 3. Compare realized return r_t to forecast VaR_t 4. Record violation indicator:

$$I_t = 1(r_t < VaR_t)$$

Backtests currently reported in the pipeline: - Kupiec unconditional coverage test - Christoffersen independence test - Conditional coverage test

Important note: - These are standard backtests and are useful. - However, for CAViaR specifically, the reference paper emphasizes the Dynamic Quantile (DQ) test as the model adequacy diagnostic. - So the current project can say the CAViaR implementation is improved and backtested, but it cannot yet claim full paper-style adequacy checking.

Interpretation: - Too many violations $\rightarrow$ model underestimates risk - Too few violations $\rightarrow$ model is conservative / overestimates risk - Failure of independence $\rightarrow$ violations cluster, model misses dynamic risk structure

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